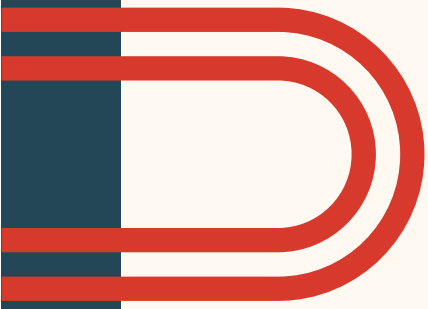




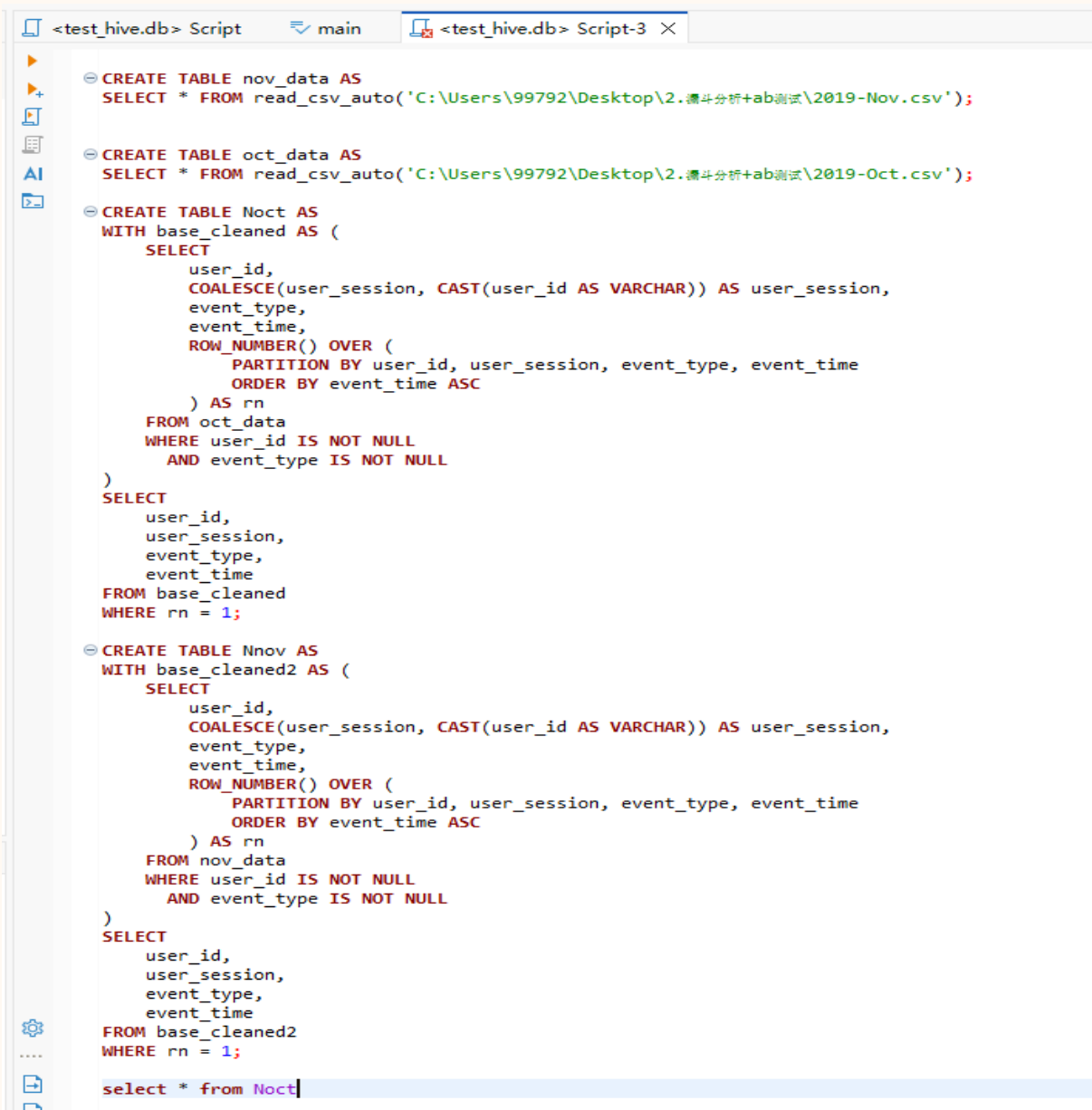
E-COMMERCE CONVERSION ANALYSIS

QINGYUE ZHANG

2026

PROJECT BACKGROUND & PROBLEM IDENTIFICATION

Objective: To drive overall transaction volume and sales growth, we conducted an end-to-end user conversion funnel analysis based on October behavior data.



```
<test_hive.db> Script    main    <test_hive.db> Script-3 X

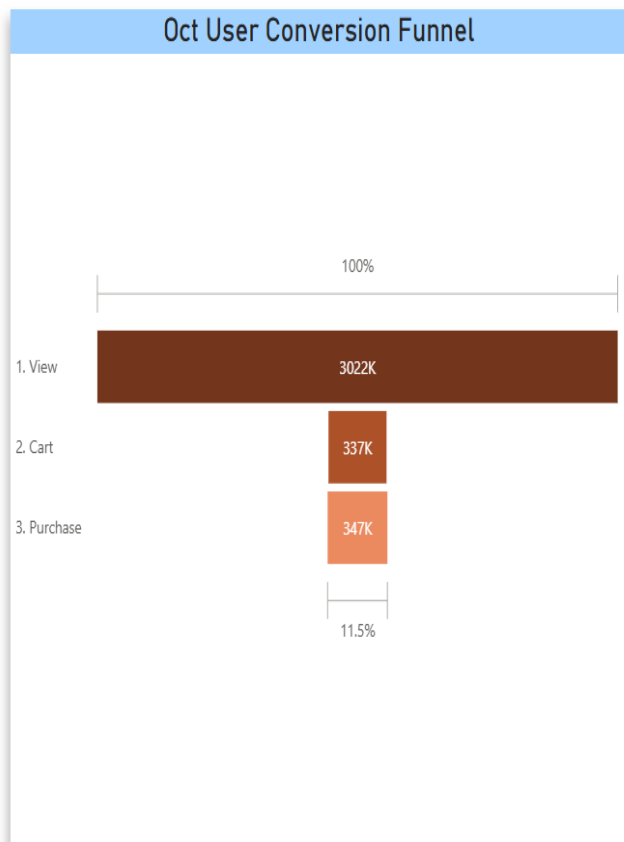
CREATE TABLE nov_data AS
SELECT * FROM read_csv_auto('C:\Users\99792\Desktop\2.漏斗分析+ab测试\2019-Nov.csv');

CREATE TABLE oct_data AS
SELECT * FROM read_csv_auto('C:\Users\99792\Desktop\2.漏斗分析+ab测试\2019-Oct.csv');

CREATE TABLE Noct AS
WITH base_cleaned AS (
    SELECT
        user_id,
        COALESCE(user_session, CAST(user_id AS VARCHAR)) AS user_session,
        event_type,
        event_time,
        ROW_NUMBER() OVER (
            PARTITION BY user_id, user_session, event_type, event_time
            ORDER BY event_time ASC
        ) AS rn
    FROM oct_data
    WHERE user_id IS NOT NULL
    AND event_type IS NOT NULL
)
SELECT
    user_id,
    user_session,
    event_type,
    event_time
FROM base_cleaned
WHERE rn = 1;

CREATE TABLE Nnov AS
WITH base_cleaned2 AS (
    SELECT
        user_id,
        COALESCE(user_session, CAST(user_id AS VARCHAR)) AS user_session,
        event_type,
        event_time,
        ROW_NUMBER() OVER (
            PARTITION BY user_id, user_session, event_type, event_time
            ORDER BY event_time ASC
        ) AS rn
    FROM nov_data
    WHERE user_id IS NOT NULL
    AND event_type IS NOT NULL
)
SELECT
    user_id,
    user_session,
    event_type,
    event_time
FROM base_cleaned2
WHERE rn = 1;

select * from Noct
```



The overall conversion rate is only 11.5%.

There is a sharp drop in conversion from "View" to "Cart" (11.15%). The conversion from "Cart" to 'Purchase' is extremely efficient (> 100%).

Recommendation: Future optimizations and A/B testing should focus 100% on the "Add to Cart" step on the product detail page.

11.15%

View to Cart Rate

102.97%

Cart to Purchase Rate

Note: Cart → Purchase conversion >100% is driven by Direct Buy (Express Checkout) paths without adding to cart, as well as cross-month purchase behaviors.

A/B TESTING EXPERIMENT PLAN

1. Objective & Hypothesis

- **Background & Bottleneck:** October data reveals a severe drop-off in the **View → Cart** funnel stage, with a low conversion rate of only **11.15%**.
- **Core Hypothesis:** Introducing a "**Highlighted Discount/Coupon Alert**" visual element near the "Add to Cart" button on the Product Detail Page (View) will reduce user decision costs and boost the Add-to-Cart conversion rate.
- **Variant Design:**
 - **Control Group (Group A):** Maintain the current Product Detail Page UI (no highlighted discount alert).
 - **Treatment Group (Group B):** Display a highlighted "Add to Cart to Enjoy Discount/Coupon" prompt on the Product Detail Page.

2. Metric Framework

Metric Type	Metric Name	Formula	Expected Outcome
Primary Metric	View → Cart Rate	Cart Users/View Users	Increase(11.15%to 12.00%)
Guardrail Metric	Refund Rate	Refund Users/ Total Purchasing Users	Remain Stable
Secondary Metric	Cart→Purchase Rate	Purchase Users/ Cart Users	Remain Stable

3. Sample Size & Duration Estimation

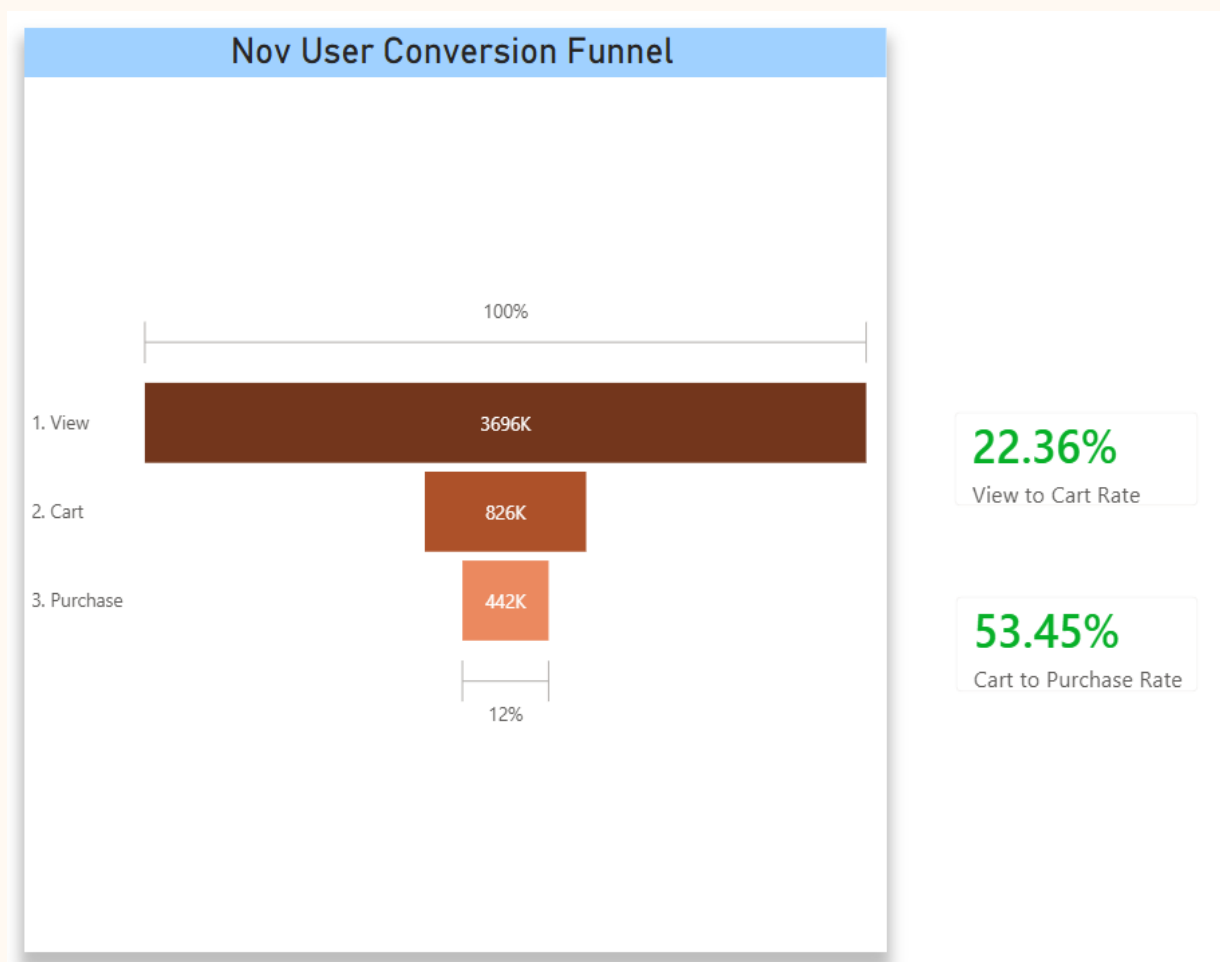
3.1 Sample Size Calculation

- **Baseline Conversion Rate (p_1):** 11.15%
- **Minimum Detectable Effect (MDE):** 0.85% (Target 12.00%)
- **Significance Level (α):** 0.05 (95% Confidence Level)
- **Statistical Power ($1 - \beta$):** 0.80 (80% Power)
- **Calculation Result:** Requires at least **22,000 UV** per group, totaling **44,000 UV** across both variants.

3.2 Traffic Allocation & Test Duration

- **Traffic Allocation:** 50% Group A : 50% Group B split.
- **Test Duration:** 14 Days.

EXPERIMENT RESULTS & STRATEGIC VERIFICATION



Note on Methodology:

Since open-source datasets lack real-time experiment tracking, November event logs were sampled and structured into a simulated 14-day 50:50 A/B test environment to evaluate the statistical framework, hypothesis testing, and guardrail check logic.

Experiment Results & Hypothesis Testing

Metric	Control Group (VariantA: Original UI)	Treatment Group (VariantB: Highlighted Coupon)	Absolute Lift	p-value	Verdict
View	1,848,000	1,848,000	-	-	50:50Traffic Split
Cart	206,052	413,212	+207,160	-	-
Primary Metric(Vie w->Cart Rate)	11.15%	22.36%	+11.21%	<0.001	Statistically Significant (Winner)

Next Steps for Guardrail Check: The primary metric showed a statistically significant improvement. Prior to a 100% full rollout, it is recommended to pull transaction database logs to verify guardrail metrics (e.g., Refund Rate, Return Rate). If no statistically significant degradation is observed, proceed with full deployment.